Manchester Metropolitan University

Department of Computing and Mathematics

ASSESSMENT COVER SHEET 2024/25

Module code and title:	6G5Z0029 Database Models and Implementation
Assignment set by:	Andrew Schofield / Anthony Kleerekoper
Assignment ID: *(e.g. 1CWK50)*	1CWK100
Assignment weighting:	100%
Assignment title:	Database Migration Project
Type: (Group/Individual)	Group
Hand-in deadline:	Fri, 16th May 2025 by 21:00
Hand-in format and mechanism:	Report submitted via Moodle

Learning outcomes being assessed:

LO1: Evaluate the suitability of various database models and architectures for a given set of data storage and processing requirements.
LO2: Describe how structured, semi-structured and unstructured data can be stored and processed by different database systems.

Note: it is your responsibility to make sure that your work is complete and available for marking by the deadline. Make sure that you have followed the submission instructions carefully, and your work is submitted in the correct format, using the correct hand-in mechanism (e.g., Moodle upload). If submitting via Moodle, you are advised to check your work after upload, to make sure it has uploaded properly. If submitting via OneDrive, ensure that your tutors have access to the work. Do not alter your work after the deadline. You should make at least one full backup copy of your work.

Penalties for late submission

The timeliness of submissions is strictly monitored and enforced.

All coursework has a late submission window of 7 calendar days, but any work submitted within the late window will be capped at 40%, unless you have an agreed extension. Work submitted after the 7-day late window will be capped at zero unless you have an agreed extension. See ‘Assessment Mitigation’ below for further information on extensions.

Please note that individual tutors are unable to grant extensions to coursework.

Assessment Mitigation

If there is a valid reason why you are unable to submit your assessment by the deadline you may apply for assessment mitigation. There are two types of mitigation you can apply for via the module area on Moodle (in the ‘Assessments’ block on the right-hand side of the page):

- **Non-evidenced extension**: does **not** require you to submit evidence. It allows you to add a **short** extension to a deadline. This is not available for event-based assessments such as in-class tests, presentations, interviews, etc. You can apply for this extension during the assessment weeks, and the request must be made **before** the submission deadline. For this assessment, the non-evidenced extension is 2 days.

- **Evidenced extension**: requires you to provide independent evidence of a situation which has impacted you. Allows you to apply for a longer extension and is available for event-based assessment such as in-class test, presentations, interviews, etc. For event-based assessments, the normal outcome is that the assessment will be deferred to the summer reassessment period.

Further information about Assessment Mitigation is available on the dedicated Assessments page.

Personal Learning Plans

If you have a Personal Learning Plan (PLP) which states you can negotiate an extended deadline, make an appointment to see the Department's Disability Coordinator to discuss to discuss your needs, and where appropriate agree on a revised submission deadline.

Plagiarism

Plagiarism is the unacknowledged representation of another person's work, or use of their ideas, as one's own. Manchester Metropolitan University takes care to detect plagiarism, employs plagiarism detection software, and imposes severe penalties, as outlined in the Student Code of Conduct and Academic Misconduct Policy. Poor referencing or submitting the wrong assignment may still be treated as plagiarism. If in doubt, seek advice from your tutor.
As part of a plagiarism check, you may be asked to attend a meeting with the Module Leader, or another member of the module delivery team, where you will be asked to explain your work (e.g. explain the code in a programming assignment). If you are called to one of these meetings, it is very important that you attend.

Use of generative AI

The use of generative AI is not permitted for this assessment. This includes but is not limited to tools like Microsoft Copilot or ChatGPT (please see What is Artificial Intelligence? if you are not sure). All submitted work must be your own original content, created without the aid of generative AI. Failure to adhere to this policy may constitute academic misconduct and result in serious consequences. If you have been advised to use a specific tool as part of a personal learning plan, you should continue to use it.

If you are unable to upload your work to Moodle

If you have problems submitting your work through Moodle, there is a Contingency Submission Form on the university's Assist ticketing system, where you can upload your work. If you use this submission method, your work must be uploaded **by the published deadline**, or it will be logged as a late submission. Alternatively, you can save your work into a single zip folder then upload the zip folder to your university OneDrive and submit a Word document to Moodle which includes a link to the folder. **It is your responsibility to make sure you share the OneDrive folder with the Module Leader, or it will not be possible to mark your work**.

Assessment Regulations

For further information see Assessment Regulations for Undergraduate/Postgraduate Programmes of Study on the Student Life web pages.

Formative Feedback:	Formative feedback will be provided during the lab sessions throughout the module by your lab tutor and during the office hours of the lecturers/lab tutors.
Summative Feedback Format:	Marks and feedback on submissions will be provided 4 weeks after the deadline including marks for each section and feedback on what was done well and what could have been improved. This will be available via Moodle.

Assignment Details and Instructions

Scenario:
A media rental company has been compiling a relational database of films and TV episodes over many years. Initially, the database was small with limited information about each title. Over time, it has grown to encompass thousands of titles and include information about each title's directors, writers, cast and crew. Unfortunately, this growth was not planned very well and the database is now a bit of a mess. There are two tables: titles and names. The titles table stores information about each title whilst the names table stores information about named individuals who are related to those titles in some way. The titles table contains 44,733 rows and the names table contains 39,070 rows.

The relational database is used to provide output to an online front-end where customers and the public can view information about titles. The database is also used to provide title recommendations to users through a combination of SQL queries and analytical algorithms.

The Project:
At present, the company manages with its two very large and poorly designed tables. The company now wants to migrate to a better design which will reduce redundancy and is also wondering if moving from a relational to a NoSQL document-based database would be advisable. The existing data will have to be transferred to the new design. An important consideration is that it must be possible to produce the same output as the existing design.

You have been contracted to redesign the relational database, create a prototype of the new relational database, transfer the data and design new queries. You will also create an additional limited prototype demonstrating how the data could be stored in a NoSQL document-based database and advise the company on the pros and cons of using each type of database system.

The project is in two parts:
- **Part A** deals with the relational database re-design, data migration and queries.
- **Part B** deals with the NoSQL re-design, migration and queries.

The project is to be conducted in groups of two or three. It is your group's responsibility to manage the project appropriately including the organization of suitable teamworking and communication methods and tools. You should stay in close contact with your group members during the project and it is recommended that you organize regular meetings outside your timetabled classes. If there are problems with the groupwork aspect of the project, such as a lack of participation from a group member, you should attempt to resolve these issues in the first instance. However, if unresolvable problems should occur within the group, you may submit the optional peer review form available on the module's Moodle page detailing the participation of each group member. If sufficient evidence is submitted, this will be considered during marking.

Project Requirements
Part A (60 marks)
Section A – Critiquing the Current Database Design (10 marks)

1. Read the scenario and project brief above and examine the three provided database dump files (db_create.sql, names_insert.sql and titles_insert.sql – these files contain SQL Create Table and Insert statements). You should take some time to try and work out what each column contains and how the columns relate to each other.

   The design is flawed which brings problems with it. You should critique the design, writing a short report outlining the mistakes in the design and what problems they lead to. This report should be approximately 500 words and should include an ERD describing the original tables as given to you in the dumps. The ERD should include full details of the relationships between the two tables, including any primary/foreign key relationships that would be needed to implement those relationships.

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Section B – New Relational Database Design (15 marks)

2. You must design an Entity Relationship Diagram with an appropriate level of normalisation (i.e. up to and including 3NF) that can store the same information as the existing tables. You are free to use either traditional "crow's foot" or UML Class Diagram notation but should include appropriate relationships/associations (including cardinality/multiplicity) between entities/classes and appropriate attributes, including primary and foreign key definitions. Your final design should resolve any many-to-many relationships.

Note: Refer to the lecture and lab notes for guidance on drawing these diagrams. You are free to use any software to draw the diagrams, but Visual Paradigm is recommended, as this will be covered in the course material.

Section C – Relational Database Implementation and Migration (20 marks)

3. **Implementation:** Implement your relational database design using SQL. You should write the statements yourself and not rely on graphical tools to do it. For this assignment, you should the MySQL/MariaDB DBMS using the University's server (Mudfoot) and the MySQL Workbench client software. You should have received an email from us with your connection details for the Mudfoot server.
Alternatively, you can use MySQL/MariaDB on your own machine via XAMPP (https://www.apachefriends.org/index.html) or by installing the MySQL Server and MySQL Workbench client.

4. **Data Migration:** Populate your new relational database with the data from the existing tables by using appropriate and correct SQL statements. This will require creating tables for the dumped data to be inserted into before the data can be inserted from them into the new tables.

Section D – SQL Query Creation (15 marks)

5. The company has a couple of complex queries that it has working with the old design. Based on your new design, you should create new queries that produce the same output. To help you, we have included a screenshot of some of the output.

   a. Count the number of people involved (in any job role) with titles of each genre as well as the total number of people involved (again in any job role) regardless of genre.

	GENRE	Number
1	Action	4451
2	Adult	10
3	Adventure	3165
4	Animation	788
5	Biography	1234
6	Comedy	11651
7	Crime	4068
8	Documentary	4241
9	Drama	18747
10	Family	1837
11	Fantasy	1326
12	Film-Noir	170
13	History	888
14	Horror	3103
15	Music	757
16	Musical	1074
17	Mystery	1803
18	News	79
19	Reality-TV	24
20	Romance	5216
21	Sci-Fi	1136
22	Sport	460
23	Thriller	3566
24	War	889
25	Western	813
26	(null)	37011

b. List all the living cinematographers who are known for titles with average ratings of 4.5 or less and the name of the title with their lowest rating. Order the results by the lowest average rating from highest to lowest.

SQL | Fetched 50 rows in 0.347 seconds

	NAME	MIN(AVERAGERATING)	TITLE
1	Evan Marlowe	4.5	Horror House
2	Dharmadjie	4.5	Hantu Jeruk Purut
3	Yuva	4.5	Jackson Durai
4	Reza Serkanian	4.5	Hip Moves
5	Jean-Max Bernard	4.5	Va mourire
6	Brian Pratt	4.5	Pauly Shore Is Dead
7	Richard King	4.5	The First Turn-On!!
8	Alain Betrancourt	4.5	Top of the World
9	Hitoshi Kato	4.5	Tomie: Revenge
10	Lachlan Milne	4.4	Uninhabited
11	Adam Sherer	4.4	Vampie
12	Goof de Koning	4.4	The Human Centipede (First Sequence)
13	Ivan Zuccon	4.4	Wrath of the Crows
14	Bryan Koss	4.4	In Stereo
15	Cuneyt Denizer	4.4	Super incir
16	Bala Bharani	4.3	Goripalayam
17	Hector Ortega	4.3	The Night Buffalo
18	Tom Adair	4.3	Complete Guide to Guys
19	Don Stern	4.3	Creature
20	Harry Mathias	4.3	Creature
21	Mark Faulkner	4.3	Complete Guide to Guys
22	Peter Roos	4.3	Suzanne og Leonard
23	Troy Smith	4.2	Time Runner
24	Oscar Lobo	4.2	Point of Contact
25	Justin McConnell	4.2	Broken Mile

Part B (40 marks)

Following the re-design and migration of the relational database, the media rental company is wondering if, in the future, they should move to a NoSQL database. They have selected the MongoDB Document-based Database and want to know how they would store and query their data and what advantages or disadvantages this approach would bring.

For this part, we have provided a smaller sample of data from the Titles and Names tables. These are available on the Moodle page as "`Part B Names.sql`" and "`Part B Titles.sql`".

Section E – NoSQL Document Database Design (25 marks)

6. Put the sample data into JSON documents, suitable for inserting into a MongoDB database. You should also write the code to perform the inserts from those documents.

You will need to decide how to structure the data and must submit a short document justifying your chosen structure. This document should be no more than 1 page (type 11 font) and should explain all non-trivial decisions in your design.

Note: You may use any tools you want to move the data into your chosen structure. This includes typing by hand, copying and pasting, using conversion tools such as https://csvjson.com/, or writing custom code in your favourite (or non-favourite) programming language.

Section F – MongoDB queries (15)

7. Write MongoDB queries to produce the following outputs, similar to those in Part A Section D. For each, we have given the output returned by the data in its current, relational-database form.

a. Count the number of people involved in each genre, in any capacity:

genre	Number
Action	63
Adventure	55
Biography	10
Comedy	41
Crime	20
Drama	160
Fantasy	20
History	10
Horror	35
Music	20
Mystery	36
Romance	34
Sci-Fi	66
Short	4
Thriller	53
War	10

b. List all the living cinematographers, who are known for at least one title with an average rating of 4.5 or less and the names of their titles with average ratings below 4.5. Order the results by the cinematographers' names.

Result Grid | Filter Rows: | Exp

name	averagerating	title
Edreace Purmul	4.4	The Playground
Jamie Bell	4.3	Fantastic Four

Deliverables

You should submit a report containing:

Part A

- Your group's ERD for the original design (section A)
- A critique of the original design (word limit about 500 words, section A)
- Your group's ERD for the new design (section B).
- The SQL code used to implement your new design (section C.3).
- The SQL code used to migrate the data including the code used to create the tables to store the dumped data (if appropriate) (section C.4)
- The SQL code for the queries (section D)

Part B

- Your group's code for storing the data about the specified titles and names in MongoDB and the justification for the chosen structure (Section E)
- Your group's code for the MongoDB queries (Section F)

Your report will be submitted via the module's Moodle page. Instructions on submission are provided on the page.

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Marking Scheme

The deliverables for the project are a group report and relevant SQL scripts.

The criteria that will be applied to each part of the submission are given in the tables below. Percentages are provided for each section to enable you to work out an overall percentage grade for the work:

Part A

Original Relational Database Design Critique (10%)	
Pass (3rd)	ERD provided with little or no critique
Lower 2nd	Mostly correct ERD provided with simple critique covering some of the issues with the design.
Upper 2nd	Correct ERD provided with good critique covering most of the issues with the design.
1st	Correct ERD provided with very good critique covering all of the issues with the design.

New Relational Database Design (15%)	
Pass (3rd)	Basic ERD produced with some normalisation but little or no supporting commentary.
Lower 2nd	ERD provided with acceptable amount of normalisation and some commentary. Appropriate use of diagram features and syntax. Attributes for diagram are mostly correct.
Upper 2nd	ERD and appropriate normalisation provided along with supporting information. Correct and accurate diagram with matching attributes datatype/length definitions and keys. Some commentary and evaluation of issues surrounding development.
1st	High quality ERD specification produced with clear, comprehensive justification and evaluation of design and alternatives, and consideration of business scenario needs. Accurate and detailed specification of attributes datatype/length definitions, keys and constraints. Design is well thought out and suitable for implementation in RDBMS.

Relational Database Implementation and Migration (20%)	
Pass (3rd)	Database tables created successfully. Implementation shows some relation to the ERD and attribute definition in terms of relationships and primary/foreign keys. Data has been transferred with some errors or workarounds.
Lower 2nd	Database implementation has been performed correctly and shows a clear relationship to the ERD in terms of relationships and primary/foreign keys. Data has been transferred with few errors or workarounds.
Upper 2nd	Properly implemented database which clearly reflects the ERD and attribute list. Primary/Foreign keys created properly and matching ERD. Data has been transferred without errors. Good, clear discussion/commentary of implementation.
1st	Well implemented database which clearly reflects the ERD and attribute list. Primary/Foreign keys created properly and matching ERD. Additional and appropriate constraints implemented. Data has been transferred using efficient SQL commands. Good, clear discussion/commentary of implementation showing consideration of business scenario requirements.

SQL Query Creation (15%)
7.5% for each correct query

Part B

NoSQL (MongoDB) Design (25%)	
Pass (3rd)	Somewhat appropriate document structure with most of the data on the given titles and names included but with some errors. Little justification given for the design which may contain poor reasoning.
Lower 2nd	Mostly appropriate document structure with most of the data on the given titles and names included but with some errors. Reasonable justification given for the design, but which may contain some poor reasoning.
Upper 2nd	Mostly appropriate document structure with all of the data on the given titles and names included but may have a small number of errors. Good justification given for the design, but which may contain the occasional poor reasoning.
1st	Fully appropriate document structure with all of the data on the given titles and names included without errors. Very good justification given for the design.

NoSQL Query Creation (15%)	
Pass (3rd)	A reasonable attempt was made to write the queries which is not far away from working correctly for the chosen document structure.
Lower 2nd	A good attempt was made to write the queries which is very close to working correctly.
Upper 2nd	A good attempt was made to write the queries which works correctly for one query is very close to working correctly for the other.
1st	Both queries work correctly.